Oracle SQL Lab Manual

Course: Diploma in Computer Science & Engineering

Submitted By:

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Roll No: ____________________ Semester: ____________

Institution Name: ___________________________________

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Exercise 1: INNER JOIN

Aim:

To display matching records from two tables using INNER JOIN.

Algorithm:

- Create Students and Courses tables.
- Use INNER JOIN to fetch only matching rows.

SQL Code:

```
SELECT S.student_id, S.name, C.course_name
FROM Students S
INNER JOIN Courses C
ON S.student_id = C.student_id;
```

Expected Output:

Displays only students who are enrolled in a course.

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Exercise 2: LEFT JOIN

Aim:

To display all records from the left table and matching records from the right using LEFT JOIN.

Algorithm:

- Use LEFT JOIN between Students and Courses tables.

SQL Code:

```
SELECT S.student_id, S.name, C.course_name
FROM Students S
LEFT JOIN Courses C
ON S.student_id = C.student_id;
```

Expected Output:

All students are listed; if a student has no course, course_name will be NULL.

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Exercise 3: RIGHT JOIN

Aim:

To display all records from the right table and matching records from the left using RIGHT JOIN.

Algorithm:

- Use RIGHT JOIN between Students and Courses tables.

SQL Code:

```
SELECT S.student_id, S.name, C.course_name
FROM Students S
RIGHT JOIN Courses C
ON S.student_id = C.student_id;
```

Expected Output:

All courses are listed; if no student is enrolled, student details will be NULL.

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Exercise 4: FULL OUTER JOIN

Aim:

To display all records from both tables using FULL OUTER JOIN.

Algorithm:

- Use FULL OUTER JOIN between Students and Courses tables.

SQL Code:

```
SELECT S.student_id, S.name, C.course_name
FROM Students S
FULL OUTER JOIN Courses C
ON S.student_id = C.student_id;
```

Expected Output:

Shows all students and all courses, with NULLs where no match exists.

Exercise 5: CROSS JOIN

Aim:

To display Cartesian product of two tables using CROSS JOIN.

Algorithm:

- Use CROSS JOIN between Students and Courses tables.

SQL Code:

```
SELECT S.name, C.course_name
FROM Students S
CROSS JOIN Courses C;
```

Expected Output:

Every student paired with every course.

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Exercise 6: SELF JOIN

Aim:

To join a table with itself using SELF JOIN.

Algorithm:

- Use aliases for the same table.
- Join on department and exclude same student_id.

SQL Code:

```
SELECT A.student_id, A.name, B.name AS Classmate
FROM Students A
JOIN Students B
ON A.dept = B.dept
AND A.student_id <> B.student_id;
```

Expected Output:

Shows students with their classmates from the same department.